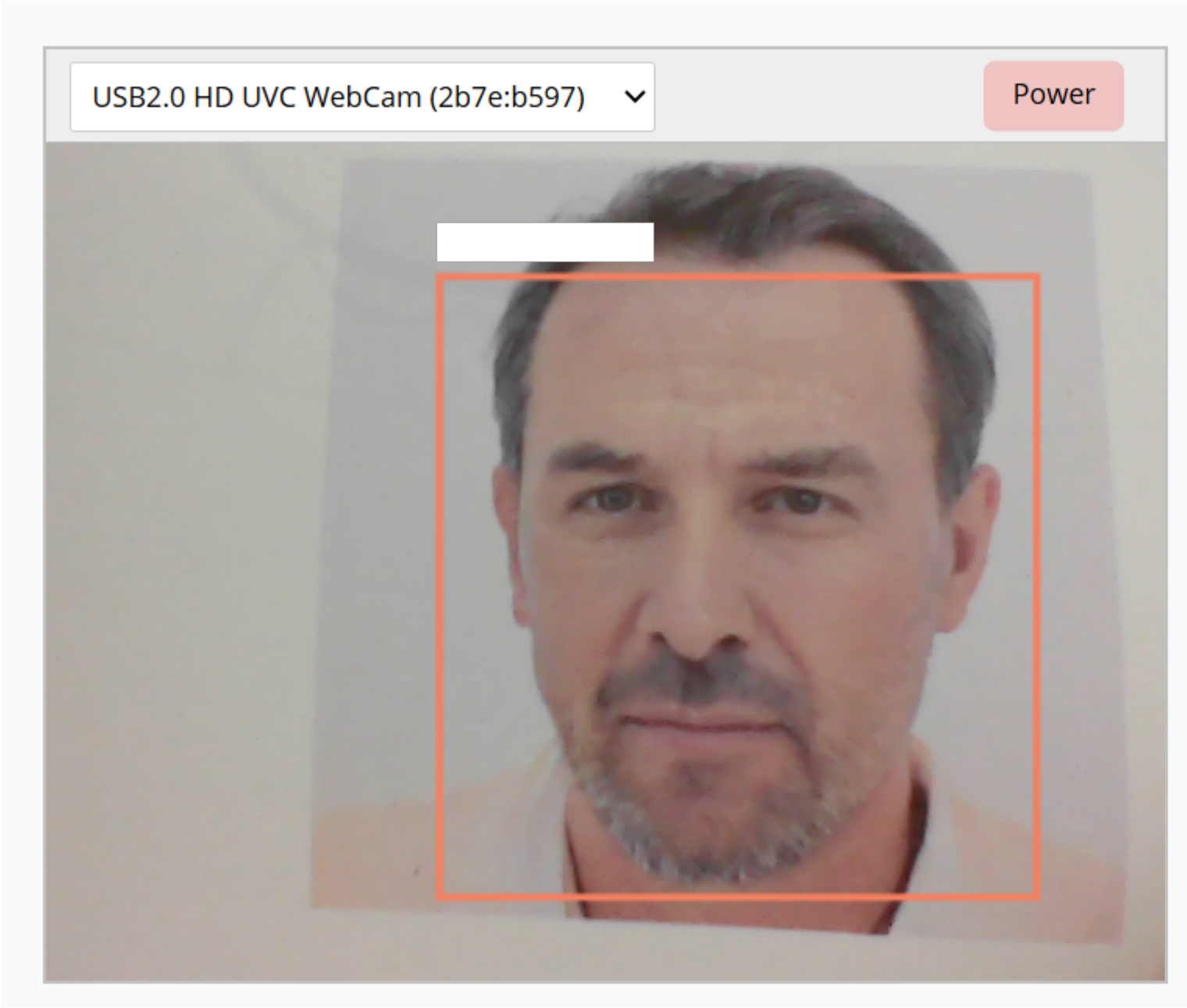
Open Dashboard

Open Dashboard is not a block that can be used in block coding,
but it allows you to open a dashboard where you can check how the model used in the extension module is applied.

Dashboard Screen

When you click Open Dashboard, you can see the following screen.



Detailed Features

Power

Turns the selected camera on or off.

Load Model

Loads the trained age, gender, expression models. This is a required step to use the 'Face Expression' extension module.

Show Result

Displays the predicted age, gender, and expression results on the camera screen.

Predict

Starts or stops age, gender, expression recognition.
You can choose whether to run it once with the Once button or continuously with the Continuous button.
You can also stop the prediction using the Stop button.

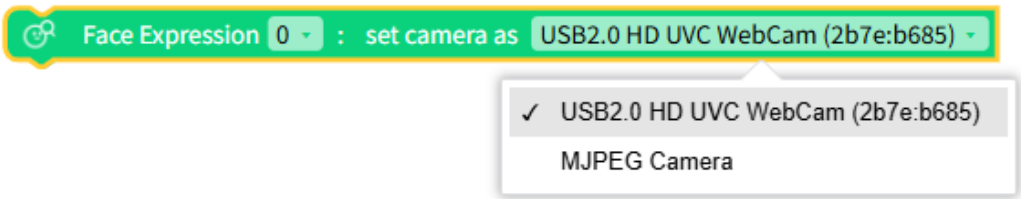
Sensory Data

Outputs the predicted age, gender, expression recognition data values.

Blocks

Set Camera

Select the camera to be used for the Face Expression module.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
camera	Dropdown option	Camera to use	List of available cameras

JavaScript Code

```
// Set a specific camera for face expression recognition (id is an example)
$('FaceExpression*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb';
```

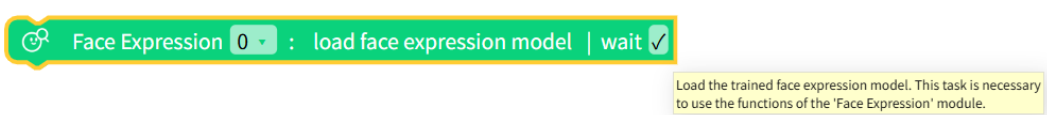
Python Code

```
# Set a specific camera for face expression recognition (id is an example)
__('FaceExpression*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb'
```

Load Face Expression Model

Loads the trained age, gender, expression model.

This step is required to use the functions of the 'Face Expression' module.



JavaScript Code

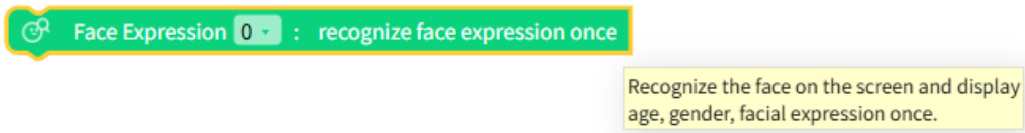
```
// Load age, gender, expression model | Wait O
$('FaceExpression*0:load_model').d = 1;
await $('FaceExpression*0:!load_model').w();

// Load age, gender, expression model | Wait X
$('FaceExpression*0:load_model').d = 1;
```

Python Code

```
# Load face model | Wait O
__('FaceExpression*0:load_model').d = 1
await __('FaceExpression*0:!load_model').w()

# Load face model | Wait X
__('FaceExpression*0:load_model').d = 1
```

Recognize Age, Gender, Expression Once

Analyzes the face currently on the screen and displays the predicted age, gender, and expression exactly once.



JavaScript Code

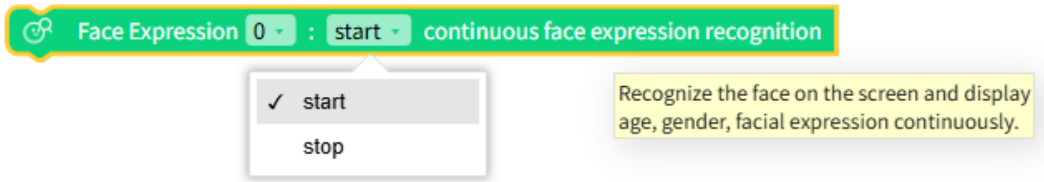
```
// Recognize age, gender, and expression once
$('FaceExpression*0:predict.once').d = 1;
```

Python Code

```
# Recognize age, gender, and expression once
__('FaceExpression*0:predict.once').d = 1
```

Recognize Age, Gender, Expression Continuously

Starts or stops continuous age, gender, expression recognition.
When continuous prediction starts, it continuously follows the face currently on the screen and displays the predicted results on the screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown option	Predict	Start (1), Stop (0)

JavaScript Code

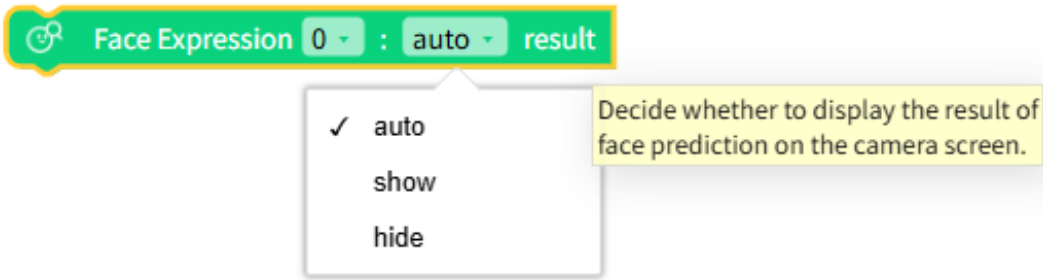
```
// Start continuous age, gender, expression recognition
$('FaceExpression*0:predict.continuous').d = 1;

// Stop continuous age, gender, expression recognition
$('FaceExpression*0:predict.continuous').d = 0;
```

Python Code

```
# Start continuous age, gender, expression recognition
__('FaceExpression*0:predict.continuous').d = 1

# Stop continuous age, gender, expression recognition
__('FaceExpression*0:predict.continuous').d = 0
```

Show Age, Gender, Expression Detection Results

Determines whether to display the predicted age, gender, and expression results on the camera screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown option	Prediction results	Show (1), Hide (0)

JavaScript Code

```
// Show age, gender, expression recognition results
$('FaceExpression*0:display').d = 1;

// Hide age, gender, expression recognition results
$('FaceExpression*0:display').d = 0;
```

Python Code

```
# Show age, gender, expression recognition results
__('FaceExpression*0:display').d = 1

# Hide age, gender, expression recognition results
__('FaceExpression*0:display').d = 0
```

Age Prediction Data

Returns the age prediction result data.

JavaScript Code

```
// Age prediction data
$('FaceExpression*0:age').d;
```

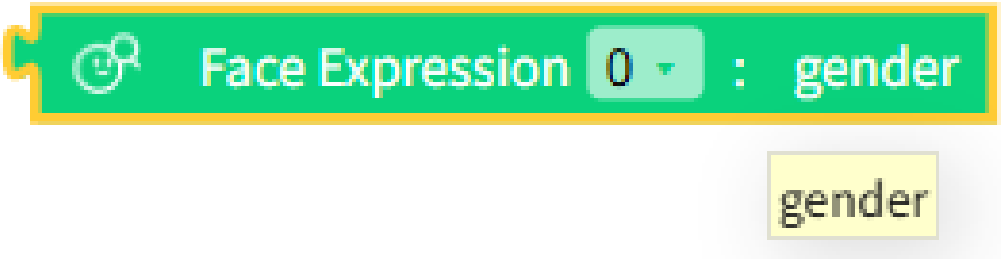
Python Code

```
# Age prediction data
__('FaceExpression*0:age').d
```

Gender Prediction Data

Returns the gender prediction result data.

The returned data is ‘Male’, ‘Female’.



JavaScript Code

```
// Gender prediction data
$('FaceExpression*0:gender.name').d;
```

Python Code

```
# Gender prediction data
__('FaceExpression*0:gender.name').d
```

Expression Prediction Data

Returns the expression prediction result data.

The returned data is ‘Happy’, ‘Sad’, ‘Angry’, ‘Afraid’, ‘Disgust’, ‘Surprise’, ‘Expressionless’.

JavaScript Code

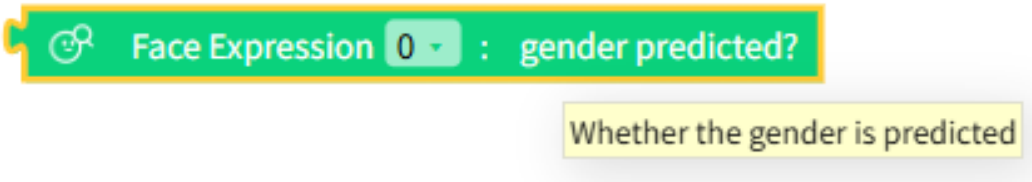
```
// Expression prediction data
$('FaceExpression*0:expression.name').d;
```

Python Code

```
# Expression prediction data
__('FaceExpression*0:expression.name').d
```

Is Gender Recognized?

Returns **True (1) / False (0)** depending on whether gender prediction has been made.



JavaScript Code

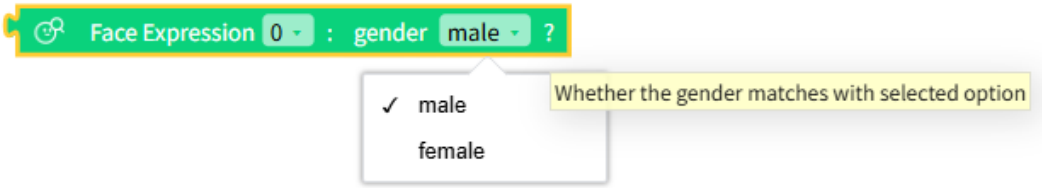
```
// Is gender recognized?
$('FaceExpression*0:gender.predicted').d;
```

Python Code

```
# Is gender recognized?
__('FaceExpression*0:gender.predicted').d
```

Is Gender ~ ?

Returns **True (1) / False (0)** depending on whether the selected gender matches the predicted gender.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
gender	Dropdown option	Gender	male, female

JavaScript Code

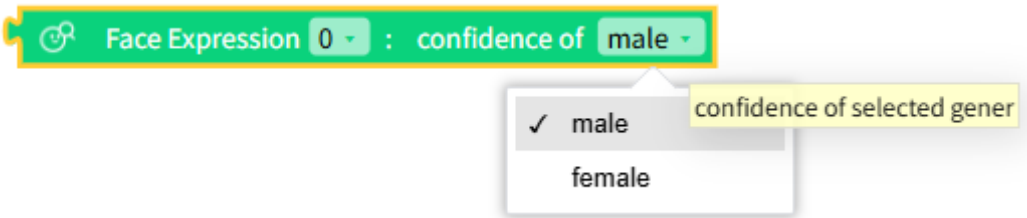
```
// Is the predicted gender male?
$('FaceExpression*0:gender.name').d === 'male';

// Is the predicted gender female?
$('FaceExpression*0:gender.name').d === 'female';
```

Python Code

```python
# Is the predicted gender male?
__('FaceExpression*0:gender.name').d == 'male'

# Is the predicted gender female?
__('FaceExpression*0:gender.name').d == 'female'
```

Probability (Confidence) that Gender is ~

Returns the probability (confidence) that the selected gender matches the predicted gender.
The return value is a real number between 0 and 1.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
gender	Dropdown option	Gender	Male(0), Female(1)

JavaScript Code

```javascript
// Probability that the predicted gender is male
$('FaceExpression*0:gender.confidence').d[0];

// Probability that the predicted gender is female
$('FaceExpression*0:gender.confidence').d[1];
```

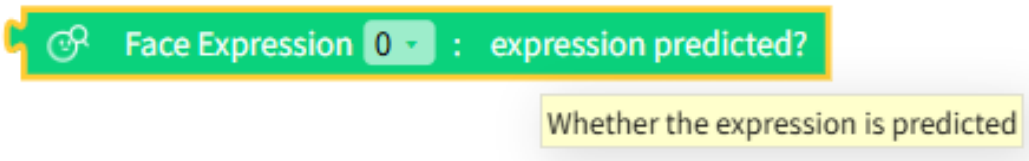
Python Code

```python
# Probability that the predicted gender is male
__('FaceExpression*0:gender.confidence').d[0]
```

```
# Probability that the predicted gender is female
__('FaceExpression*0:gender.confidence').d[1]
```

Is Expression Recognized?

Returns **True (1) / False (0)** depending on whether expression recognition has been made.



JavaScript Code

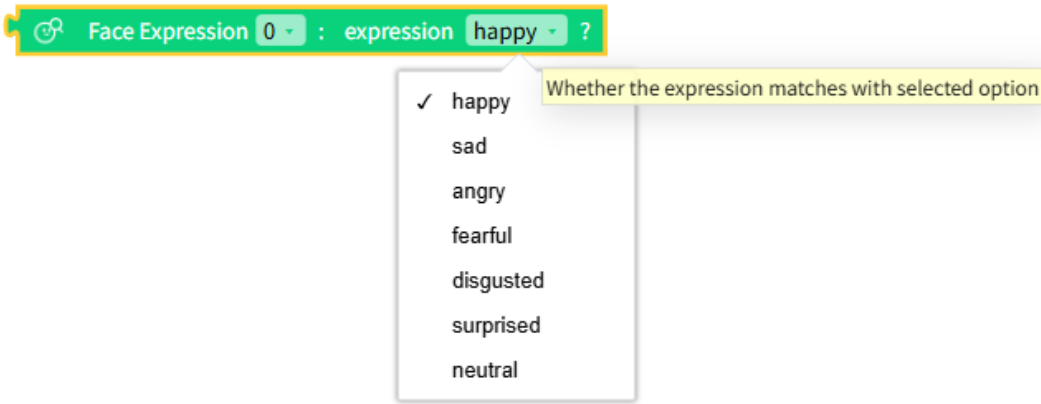
```
// Is expression recognized?
$('FaceExpression*0:expression.predicted').d;
```

Python Code

```
# Is expression recognized?
__('FaceExpression*0:expression.predicted').d
```

Is Expression ~ ?

Returns **True (1) / False (0)** depending on whether the selected expression matches the predicted expression.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
expression	Dropdown option	Expression	happy, sad, angry, fearful, disgusted, surprised, natural

JavaScript Code

```
// Is the predicted expression happy?
$('FaceExpression*0:expression.name').d === 'happy';

// Is the predicted expression sad?
$('FaceExpression*0:expression.name').d === 'sad';

// Is the predicted expression angry?
$('FaceExpression*0:expression.name').d === 'angry';

// Is the predicted expression fearful?
$('FaceExpression*0:expression.name').d === 'fearful';

// Is the predicted expression disgusted?
$('FaceExpression*0:expression.name').d === 'disgusted';

// Is the predicted expression surprised?
$('FaceExpression*0:expression.name').d === 'surprised';

// Is the predicted expression natural?
$('FaceExpression*0:expression.name').d === 'natural';
```

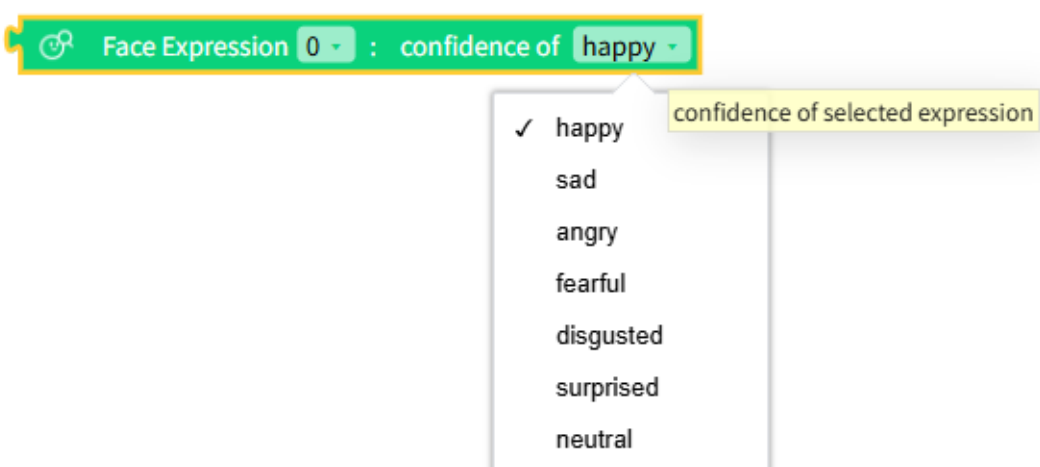
Python Code

```
# Is the predicted expression happy?
__('FaceExpression*0:expression.name').d == 'happy'
```

```
# Is the predicted expression sad?
__('FaceExpression*0:expression.name').d == 'sad'

# Is the predicted expression angry?
__('FaceExpression*0:expression.name').d == 'angry'

# Is the predicted expression fearful?
__('FaceExpression*0:expression.name').d == 'fearful'

# Is the predicted expression disgusted?
__('FaceExpression*0:expression.name').d == 'disgusted'

# Is the predicted expression surprised?
__('FaceExpression*0:expression.name').d == 'surprised'

# Is the predicted expression natural?
__('FaceExpression*0:expression.name').d == 'natural'
```

Probability (Confidence) that Expression is ~

Returns the probability (confidence) that the selected expression matches the predicted expression. The return value is a real number between 0 and 1.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
expression	Dropdown option	Expression	Happy(0), Sad(1), Angry(2), Afraid(3), Disgust(4), Surprise(5), Expressionless(6)

JavaScript Code

```
// Probability that the predicted expression is happy
$('FaceExpression*0:expression.confidence').d[0];

// Probability that the predicted expression is sad
$('FaceExpression*0:expression.confidence').d[1];

// Probability that the predicted expression is angry
$('FaceExpression*0:expression.confidence').d[2];

// Probability that the predicted expression is afraid
$('FaceExpression*0:expression.confidence').d[3];

// Probability that the predicted expression is disgust
$('FaceExpression*0:expression.confidence').d[4];

// Probability that the predicted expression is surprise
$('FaceExpression*0:expression.confidence').d[5];

// Probability that the predicted expression is expressionless
$('FaceExpression*0:expression.confidence').d[6];
```

Python Code

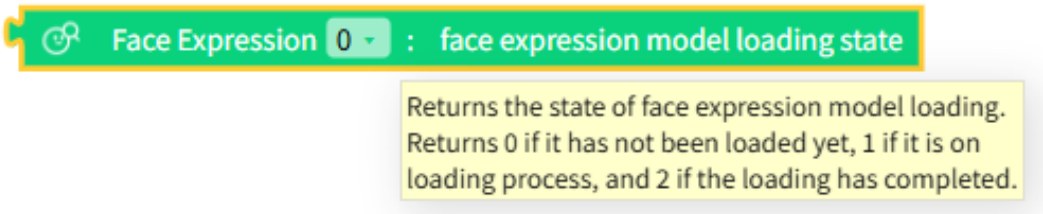
```
# Probability that the predicted expression is happy
__('FaceExpression*0:expression.confidence').d[0]
```

```
# Probability that the predicted expression is sad
__('FaceExpression*0:expression.confidence').d[1]

# Probability that the predicted expression is angry
__('FaceExpression*0:expression.confidence').d[2]

# Probability that the predicted expression is afraid
__('FaceExpression*0:expression.confidence').d[3]

# Probability that the predicted expression is disgust
__('FaceExpression*0:expression.confidence').d[4]

# Probability that the predicted expression is surprise
__('FaceExpression*0:expression.confidence').d[5]

# Probability that the predicted expression is expressionless
__('FaceExpression*0:expression.confidence').d[6]
```

Age, Gender, Expression Model Loading State Value

Returns the loading state of the age, gender, and expression model.

If it has not been loaded yet, it returns 0; if it is loading, it returns 1; if loading is complete, it returns 2.



JavaScript Code

```
// Age, gender, and expression model loading state value
$('FaceExpression*0:model_state').d;
```

Python Code

```
# Age, gender, and expression model loading state value
__('FaceExpression*0:model_state').d
```